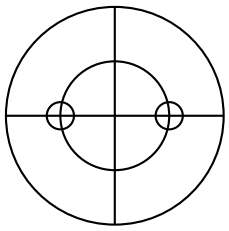
Dutton's Unified SEFI

0.5cm]**Single-Entity Field Interpretation**

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Author: Jason Dutton

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Submitted to:

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Date: August 2026

Abstract

Dutton's Unified SEFI proposes that all electrons and positrons are manifestations of a single underlying entity whose worldline spans all possible spacetime paths. The dense superposition of this worldline appears as a continuous field, reproducing the electron field of quantum field theory. Quantum interference arises from overlapping contributions of different segments of the entity's path structure, while measurement corresponds to selecting a subset of paths, producing a localized particle event. The model is formulated using relativistic worldline actions, Grassmann spin variables, electromagnetic gauge coupling, presence-density mapping, and the Dirac field in curved spacetime. SEFI preserves all predictions of quantum mechanics and quantum field theory while offering a unified ontological interpretation of particle identity.

1 Introduction

Quantum mechanics and quantum field theory describe electrons as excitations of a field, yet the ontological nature of particle identity and self-interference remains conceptually unresolved. Dutton's Unified SEFI provides a unified interpretation in which all electrons and positrons are manifestations of a single underlying entity.

2 Core Postulates

- P1 — Single underlying entity.
- P2 — Forward-time segments appear as electrons; backward-time segments appear as positrons.
- P3 — Dense spacetime weaving produces field appearance.
- P4 — Presence density corresponds to the electron field.
- P5 — Interference arises from path overlap.
- P6 — Measurement selects a subset of paths.
- P7 — Particle identity arises from ontological unity.

3 Mathematical Framework

3.1 Worldline Action

$$S_{\text{WL}} = -m \int d\tau \sqrt{-g_{\mu\nu}\dot{\gamma}^\mu \dot{\gamma}^\nu}$$

3.2 Spin Structure

$$S_{\text{spin}} = \int d\tau \left[\frac{i}{2} \Psi_a \dot{\Psi}^a \right]$$

3.3 Gauge Coupling

$$S_{\text{EM}} = q \int d\tau \, \dot{\gamma}^\mu A_\mu$$

3.4 Presence Density

$$\rho(x) = \int \mathcal{D}[\gamma] e^{\frac{i}{\hbar} S[\gamma]} \delta^{(4)}(x - \gamma(\tau))$$

3.5 Dirac Field

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$$

4 Interpretation

The field is the visible projection of one entity's full path structure. Interference arises from overlapping path contributions. Measurement corresponds to selecting a subset of paths.

5 Speculative Extensions

Golden-ratio path weighting, unified substrate interpretation, geometric self-similarity.

6 Research Note Version

Summary

SEFI interprets electrons and positrons as manifestations of a single underlying entity whose worldline spans all possible spacetime paths.

Key Equations

$$\rho(x) = \int \mathcal{D}[\gamma] e^{\frac{i}{\hbar} S[\gamma]} \delta(x - \gamma)$$

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$$

7 Conference Abstract

Dutton's Unified SEFI proposes that all electrons and positrons are manifestations of a single underlying entity whose worldline spans all possible spacetime paths. The dense superposition of this worldline appears as the electron field, while interference arises from overlapping path contributions.

8 Preprint Version

Abstract

Same as main abstract.

8.1 Condensed Introduction

SEFI provides a worldline-based interpretation of electron identity and interference.

9 Appendix A: Full Mathematical Derivations

9.1 Worldline Formalism

Detailed expansions...

9.2 Spinor Construction

Detailed derivations...

10 Appendix B: Speculative Extensions

10.1 Golden Ratio Weighting

$$w_\varphi[\gamma] = e^{\frac{i}{\hbar}S[\gamma]} \cdot f_\varphi[\gamma]$$

11 Appendix C: Superposition in SEFI — Deep Interpretation

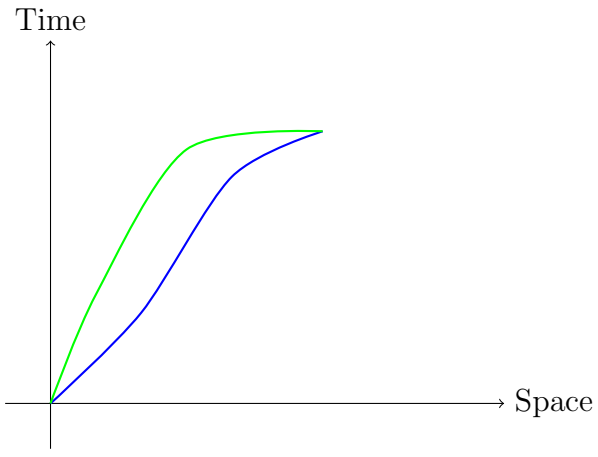
11.1 Overview

Superposition is the geometric consequence of a single entity whose worldline spans multiple allowed spacetime trajectories.

11.2 Worldline Multiplicity

$$\rho(x) = \int \mathcal{D}[\gamma]\, e^{\frac{i}{\hbar}S[\gamma]}\, \delta^{(4)}(x - \gamma(\tau)).$$

11.3 Geometric Interpretation

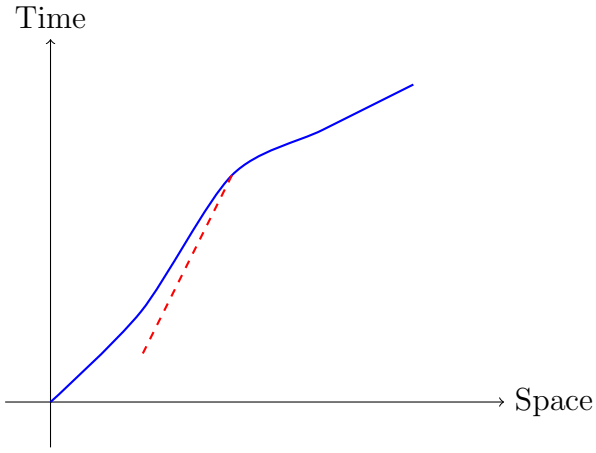


11.4 Collapse

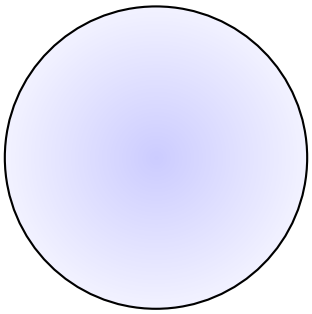
Measurement selects one worldline segment.

12 Diagram Set

12.1 Diagram 1: Worldline



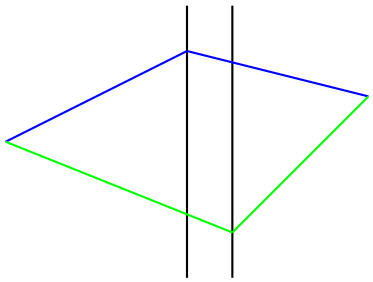
12.2 Diagram 2: Presence Density



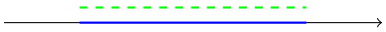
12.3 Diagram 3: Time Reversal



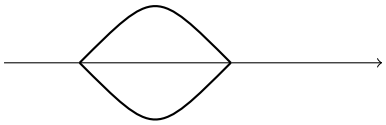
12.4 Diagram 4: Double Slit



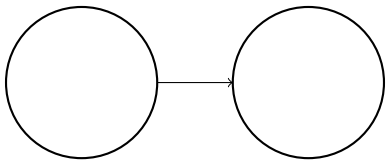
12.5 Diagram 5: Collapse



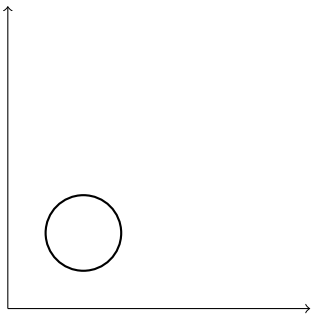
12.6 Diagram 6: Path Integral



12.7 Diagram 7: Identity Preservation



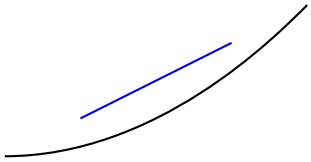
12.8 Diagram 8: Spin Structure



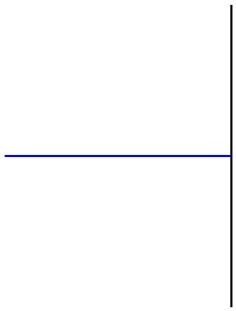
12.9 Diagram 9: Gauge Coupling



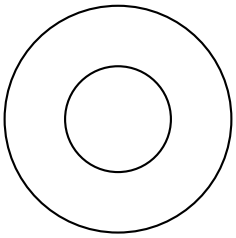
12.10 Diagram 10: Curved Spacetime



12.11 Diagram 11: Measurement Boundary



12.12 Diagram 12: Unified Entity



13 References

- Feynman, R. P. (1948). Space-time approach to quantum mechanics.
- Schwinger, J. (1951). On gauge invariance and vacuum polarization.
- Polyakov, A. M. (1987). Gauge Fields and Strings.
- Weinberg, S. (1995). The Quantum Theory of Fields.